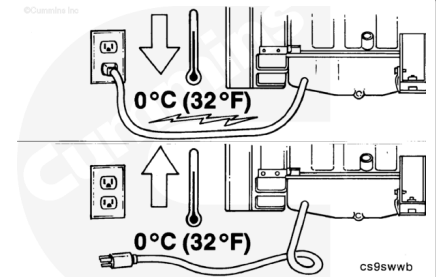


Trainings ▾

General Information

Some heaters will operate continuously when plugged into the correct voltage electrical socket. Operate them **only** when the ambient temperature is below 0°C [32°F].

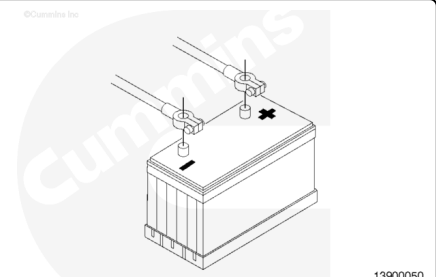


Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

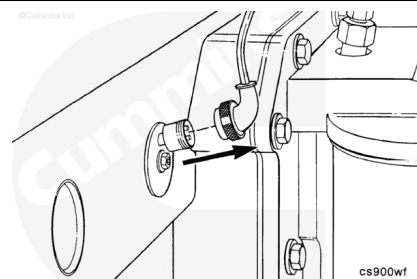
- Drain at least 19 liters [5 US Gal]. Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/40/40-008-018-tr.html).



Remove

Front Gear Train

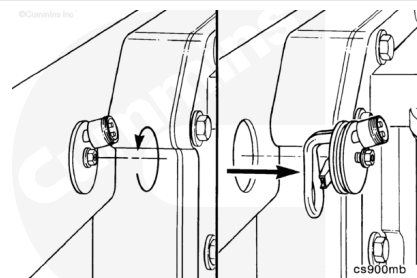
Disconnect the block heater electrical cord.



Note : Depending on the type of cylinder block, there are two types of coolant heaters used. A bolted coolant heater and a threaded coolant heater. Both are located adjacent to the lubricating oil cooler.

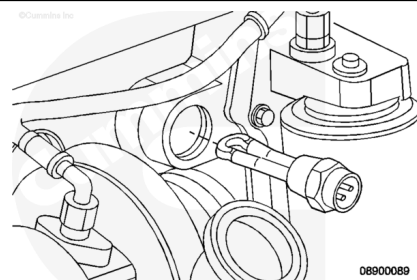
For bolted coolant heaters, loosen the block heater retaining nut.

Remove the coolant heater from the block.



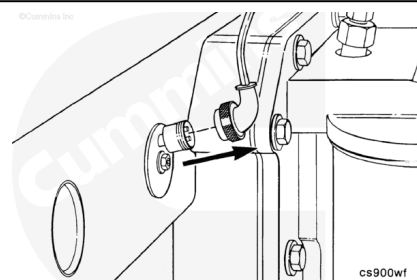
For threaded coolant heaters, unthread the coolant heater from the block.

Remove the coolant heater from the block.



Rear Gear Train

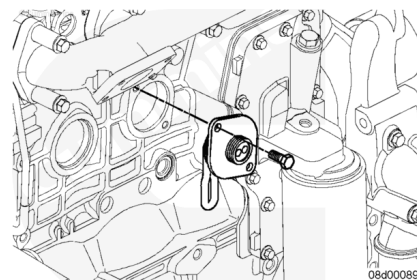
Disconnect the block heater electrical cord.



Note : Depending on the type of cylinder block, there are two types of coolant heaters used. A flange mounted coolant heater and a threaded coolant heater.

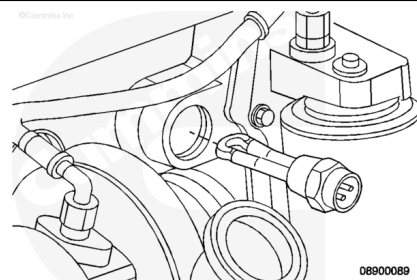
For flange mount coolant heaters, remove the two retaining capscrews.

Remove the coolant heater from the block.



For threaded coolant heaters, unthread the coolant heater from the block.

Remove the coolant heater from the block.

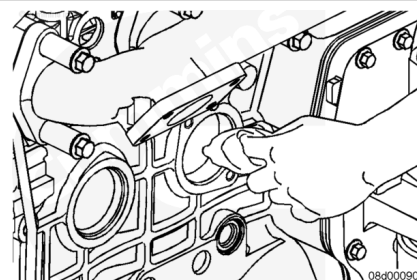


Clean and Inspect for Reuse

Clean the cup plug bore (or hole) thoroughly with a clean rag.

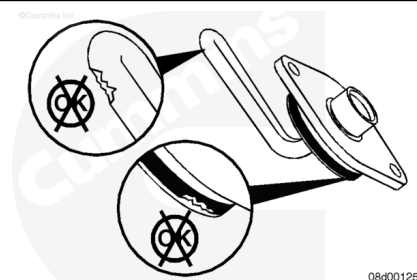
Make sure there are no burrs, metal shavings, or sharp edges that can possibly cut the o-ring.

Clean the coolant heater thoroughly with a clean rag. Make sure the heating element is free of debris and buildup and the sealing area is clean.



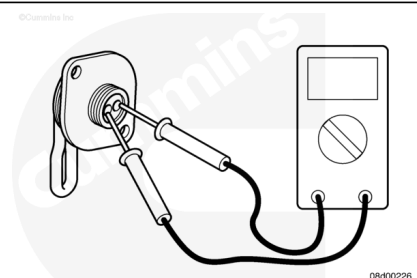
Check the coolant heater for cracks on the element.

Check the o-ring for cracks.



⚠ WARNING ⚠

To reduce the possibility of personal injury, due to electrical shock, do not touch the electrical supply wires or component while the testing procedure is in action.



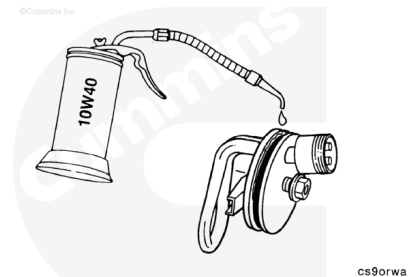
Test the coolant heater resistance. The resistance **must** read between minimum 18.2 to maximum 21.1 ohms.

Coolant Heater Resistance (Ohms)	
MIN	MAX
18.2 Ohms	21.1 Ohms

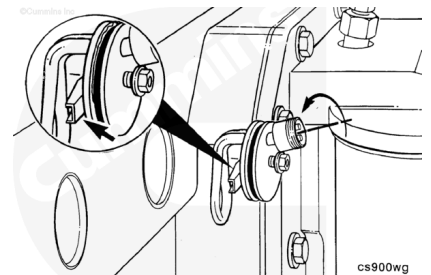
Install

Front Gear Train

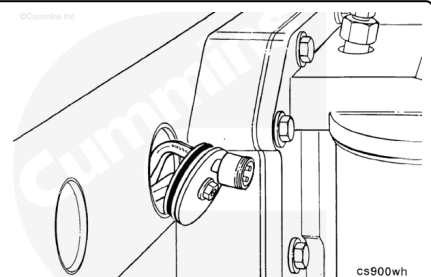
For bolted coolant heaters, lubricate the new heater o-ring with clean lubricating engine oil.



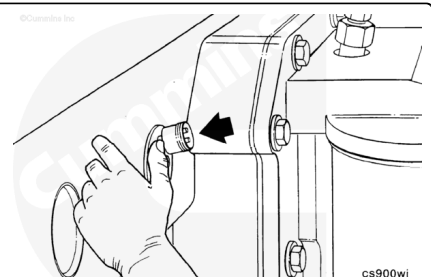
The locking channel (T-bar) should be threaded out to the end of the bolt. If so equipped, do **not** remove the retaining wire used to position the channel (T-bar).



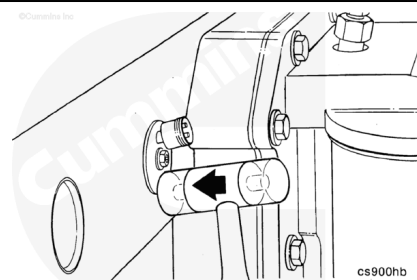
Hook the element and one leg of the channel (T-bar) into the hole as illustrated.



Hook the other leg of the channel in the hole, and push the heater into the hole as far as possible by hand.

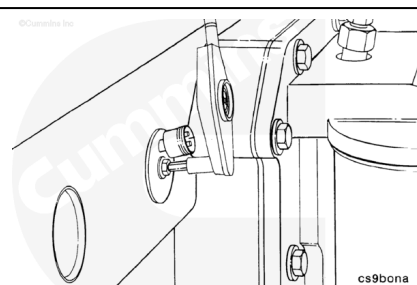


If necessary, use a plastic hammer to tap the heater in until the shoulder contacts the block.



Minimum:	1.3 n.m	[12 in-lb]
Maximum:	2.8 n.m	[25 in-lb]

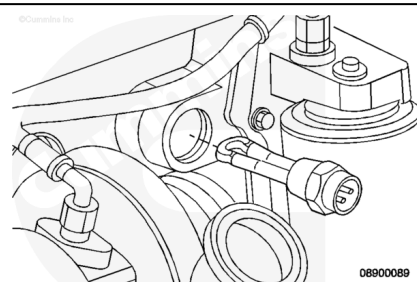
Note : Do not overtighten.



For threaded coolant heaters, apply pipe sealant, Part Number 3375066, to threads of the coolant heater.

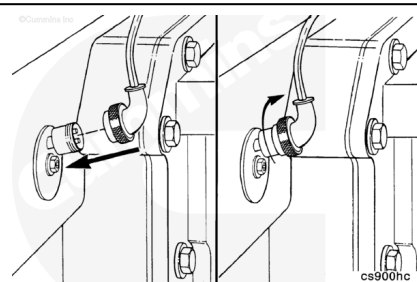
Install the coolant heater into the cylinder block and tighten.

Torque Value: 55 n•m [41 ft-lb]



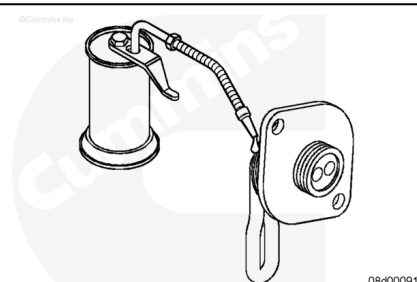
Insert the power cord into the socket being careful to align the pins with the sockets of the power cord. Tighten the retaining nut by hand.

Note : Do not apply power until the cooling system is filled, and has run long enough for the thermostat to open and allow the air to escape.



Rear Gear Train

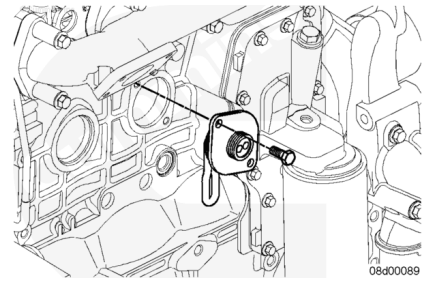
For flange mounted coolant heaters, lubricate the new heater o-ring with clean 15W-40 engine oil.



Install the coolant heater and retaining capscrews.

Tighten the two retaining capscrews.

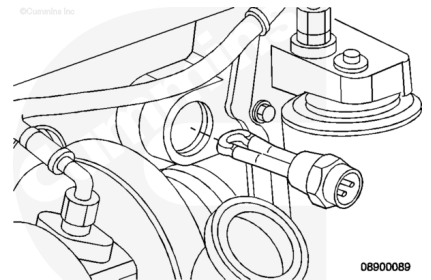
Torque Value: 24 n•m [212 in-lb]



For threaded coolant heaters, apply pipe sealant, Part Number 3375066, to threads of the coolant heater.

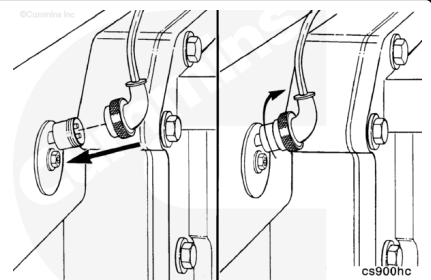
Install the coolant heater into the cylinder block and tighten.

Torque Value: 55 n•m [41 ft-lb]



Insert the power cord into the socket being careful to align the pins with the sockets of the power cord. Tighten the retaining nut by hand.

Note : Do not apply power until the cooling system is filled, and has run long enough for the thermostat to open and allow the air to escape.



Finishing Steps

⚠ WARNING ⚠

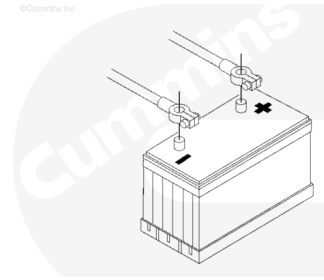
Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Fill the engine cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/40/40-008-018-tr.html>).



⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



13900050

- Connect batteries
- Operate engine and check for leaks.

Last Modified: 24-Apr-2006
